



## **Cloud Kitchen Management App Specification Document**

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## 1. Introduction:

"Ghost kitchens," also known as virtual kitchens, cloud kitchens, or dark kitchens, are commercial food preparation facilities that are designed for delivery-only meals. Unlike traditional restaurants, ghost kitchens typically don't have a physical dining area for customers. Instead, they operate solely for online orders and food delivery platforms.

Ghost kitchens have become increasingly popular, especially with the rise of food delivery services and the changing consumer habits towards convenience. They offer an opportunity for entrepreneurs to enter the food industry with lower upfront costs and less risk compared to opening a traditional restaurant.



The Cloud Kitchen Management App is a software solution aimed at optimizing inventory management and delivery operations for ghost kitchens. The app will streamline the process of tracking inventory levels, managing orders, coordinating deliveries, and analyzing data to improve efficiency and customer satisfaction.

## **2. User Story:**

### **Title:**

As a Restaurant Manager, I want to track inventory levels to ensure sufficient stock for daily operations.

### **Story:**

As a Restaurant Manager operating a busy ghost kitchen, I need a reliable inventory management system to monitor and manage inventory levels effectively.

As a user, I can access the inventory management system through a web-based interface. Upon logging in, I can view a dashboard displaying real-time inventory levels for all stocked items. I am able to search for specific inventory items by name, SKU, or category. The system provides alerts and notifications when inventory levels fall below predefined thresholds, allowing timely reordering. I can initiate purchase orders directly from the system to replenish low-stock items. The system updates inventory levels automatically upon receipt of new stock, ensuring accuracy and real-time visibility. I have access to detailed reports on inventory usage, consumption trends, and costs for analysis and planning purposes. The inventory management system integrates seamlessly with other modules of the cloud kitchen management platform, such as order processing and menu management.

### **Tasks:**

1. Design the user interface for the inventory management dashboard.
2. Implement inventory tracking functionality, including real-time updates and threshold notifications.
3. Develop the ability to generate purchase orders and track incoming stock.
4. Integrate inventory reports and analytics features into the system.
5. Conduct user acceptance testing to validate the functionality and usability of the inventory management system.

## **3. Functional Requirements:**

### **3.1 Inventory Management:**

- **Inventory Tracking:** The app should allow ghost kitchens to input and track inventory items, including ingredients, packaging materials, and other supplies.
- **Real-time Updates:** Inventory levels should be updated in real-time to reflect changes such as usage, deliveries, and new stock arrivals.
- **Automated Alerts:** The app should provide alerts and notifications for low inventory levels, enabling timely reordering to prevent stock outs.
- **Inventory Reports:** Generate detailed reports on inventory usage, wastage, and costs for analysis and optimization.

### **3.2 Order Management:**

- **Order Processing:** Receive and process incoming orders from various online delivery platforms seamlessly within the app.
- **Menu Management:** Flexible menu configuration for each brand, including item selection, pricing, and customization options. Dynamic menu updates to reflect changes in inventory availability or seasonal offerings
- **Order Status Tracking:** Track the status of orders from placement to fulfillment, including preparation, packaging, and delivery stages.
- **Order Prioritization:** Allow ghost kitchens to prioritize orders based on factors such as delivery time, customer location, and order volume.

- **Order History:** Maintain a history of past orders for reference and analysis purposes.

### **3.3 Delivery Coordination:**

- **Delivery Scheduling:** Enable ghost kitchens to schedule delivery times based on order volume and driver availability.
- **Driver Assignment:** Assign delivery orders to available drivers efficiently, considering factors like proximity to the delivery location and driver capacity.
- **Real-time Tracking:** Provide real-time tracking of delivery drivers to monitor their progress and estimated arrival times.
- **Delivery Confirmation:** Allow drivers to confirm successful deliveries and provide feedback on any issues encountered during the delivery process.

### **3.4 Analytics and Reporting:**

- **Performance Metrics:** Generate analytics and reports on key performance metrics such as order volume, delivery times, inventory turnover, and customer satisfaction ratings.
- **Trend Analysis:** Identify trends and patterns in order and inventory data to make informed decisions and improve operational efficiency.
- **Customizable Dashboards:** Provide customizable dashboards for users to view relevant metrics and data visualizations based on their roles and preferences.

## **4. Non-functional Requirements:**

- **Security:** Implement robust security measures to protect sensitive data, including encryption of user information and secure authentication mechanisms.
- **Scalability:** Design the app to handle increasing volumes of orders and inventory items as the business grows.
- **Reliability:** Ensure high availability and reliability of the app to minimize downtime and disruptions to operations.
- **Usability:** Create an intuitive and user-friendly interface to facilitate easy navigation and usage for kitchen staff, managers, and delivery drivers.

## 5. Technical Specifications:

- **Platform:** Develop the app as a cross-platform solution compatible with both iOS and Android devices.
- **Backend:** Utilize a scalable and robust backend infrastructure to handle data storage, processing, and API integrations with third-party delivery platforms.
- **Technologies:** Use modern technologies and frameworks such as React Native for the frontend, Node.js for the backend, and MongoDB for data storage.
- **API Integrations:** Integrate APIs from popular food delivery platforms (e.g., Uber Eats, DoorDash, Food Panda) for seamless order processing and delivery coordination.

## 6. Testing Requirements:

- **Unit Testing:** Perform unit tests to verify the functionality and behavior of individual app components.
- **Integration Testing:** Conduct integration tests to ensure smooth communication and interaction between different modules and third-party APIs.

- **User Acceptance Testing (UAT):** Engage ghost kitchen staff and stakeholders in UAT to validate the app's usability, functionality, and performance in real-world scenarios.

## **7. Deployment and Support:**

- **Deployment:** Deploy the app to app stores (Apple App Store, Google Play Store) and provide instructions for installation and setup.
- **Ongoing Support:** Offer ongoing technical support and maintenance services to address any issues, bugs, or feature requests that arise post-launch.

## **8. Conclusion:**

The Cloud Kitchen Management App aims to revolutionize inventory management and delivery operations for ghost kitchens, enhancing efficiency, productivity, and customer satisfaction. The app will deliver a robust, scalable, and user-friendly solution tailored to the unique needs of ghost kitchen businesses.